ONLINE REGISTRATION SYSTEM (ORS)

COMPREHENSIVE TECHNICAL EPROJECT REPORT

Inspired by Shonda Rhimes' Cinematic Medical Universe: A Visual Architecture of "Dark and Twisty" Isolation Balanced with "Bright and Shiny" Clinical Exposure.

ENGINE STACK:	LAMP Framework Architecture (Linux / Apache / MySQL / PHP / Perl)
HARDWARE THRESHOLD:	Intel Pentium 166 MHz Base | 128 MB Server Physical RAM Allocation
THEME CONTEXT:	Clinical Contrast Model - Optimized High-Contrast Low-Footprint UI Layout
DEPLOYMENT LOG:	May 2026 // Production Release Master Sign-Off Dossier

1.0 ACKNOWLEDGEMENTS

This technical portfolio and application matrix represents a rigorous implementation of lightweight server software designed for emergency health resource allocation. We express our structural appreciation to the Academic Review Committee and Computer Science Department for setting meticulous software documentation guidelines.

We explicitly acknowledge the unique visual blueprint behind this architecture, adapted from the creative concept of *Grey's Anatomy*. Balancing the moody, deep shadows of the patient's individual space (the **"Dark and Twisty"** mindset) with the hyper-exposure clarity of active emergency triage boards (the **"Bright and Shiny"** medical workspace) allowed us to manufacture an interface that decreases visual burnout during late-night tracking or quarantine containment cycles.

Finally, we acknowledge the architectural backbones of the open-source community—specifically the developers behind Apache, native procedural MySQLi connections, and Perl scripting interpreters. Their commitment to legacy runtime backwards-compatibility enables this platform to function without performance degradation inside strict vintage compute hardware limits.

2.0 ePROJECT SYNOPSIS

2.1 Executive Summary & Problem Scope

The global impact of pandemic events highlights severe distribution errors across disconnected regional healthcare systems. Lack of transparent communication between isolate consumers, local clinics, and centralized tracking authorities frequently causes queue overload and structural gridlock.

The **Online Registration System (ORS)** constructs a secure network mapping interface bridging these groups. This lightweight portal allows patients to securely self-register using mobile indices, schedule medical evaluations or immunizations, track dynamic diagnostic lab outcomes, and monitor local resource stocks.

2.2 Aesthetic Paradigm: Clinical Contrast Matrix

> *"Meredith: I'm dark and twisty... Cristina: We are dark and twisty. But it's what makes our incisions clean."*
> *— The functional design mimics this contrast: dark modes safeguard patient visibility during high-latency home isolation; bright, shadowless workspaces force high-throughput processing in clinical nodes.*

- **Patient Workspace (Dark & Twisty Mode):** Dominated by obsidian blue and dark slate styles (`#090d16`). Lowers screen glare for patients tracking symptoms in dark quarantine quarters or late-night monitoring scenarios. Status items utilize glowing high-contrast neon parameters.
- **Hospital Queue Board (Bright & Shiny Mode):** Switches completely to hard contrast light layouts (`#f8fafc`). Simulates an open operating field where zero shadow interference is permitted, favoring immediate triage decisions.

- **Admin Command Matrix (The Chief of Surgery Layout):** A centralized grid panel fusing structural navigation tools with tabular binary spreadsheet streaming configurations.

3.0 ePROJECT ANALYSIS

3.1 Feasibility Assessment Metrics

The software design prioritizes structural optimization, rendering pages entirely on the server-side to avoid processing overhead on old clients. Hardware verification tests confirm reliable compilation under an Intel Pentium 166 system with only 128 MB of server memory.

3.2 Technical Requirements Matrix

Domain Group	Explicit Technical Operational Parameter Baseline
Authentication	Encryption mapping of access tokens using one-way cryptographic algorithms (`PASSWORD_BCRYPT`).
Data Sanitization	Procedural escape sequences (`mysqli_real_escape_string`) applied globally across user forms to block injection paths.
Memory Boundary	Tabular exports stream data line-by-line using content-header overrides to fit inside the 128 MB RAM envelope.
Automation	Perl interpreter engine routines (`DBI:mysql`) monitoring biological inventory stock thresholds via system cron processes.

4.0 ePROJECT DESIGN

4.1 Data Flow Diagrams (DFDs)

Level 0: System Context Mapping

```
[ Patient User ] --(Reg Forms & Booking Entries)--> [ ORS BACKEND SYSTEM ] [ Patient User ] <--(Lab Results & Schedule Notes)-- [ ORS BACKEND SYSTEM ] [ Hospital Node ] <--(Triage Approvals & Updates)--- [ ORS BACKEND SYSTEM ] [ Admin Command ] --(Global Verification Sweeps)--> [ ORS BACKEND SYSTEM ]
```

Level 1: System Process Decomposition Architecture

```
[ Patient Input ] -> (P-1.0: Cryptographic Login) -> [ Data Layer: PATIENT TABLE ] [ Booking Forms ] -> (P-2.0: Triage Queue Insertion) -> [ Data Layer: APPOINTMENT TABLE ] [ Clinical Desk ] -> (P-3.0: Diagnostic Result Mutation) -> [ Data Layer: APPOINTMENT TABLE ] [ Date Triggers ] -> (P-4.0: XLS Spreadsheet Streaming) -> [ Binary Output File ]
```

4.2 Logical Flowcharts

Patient Booking Validation Life-Cycle

```
[ START ] -> [ Load patient_portal.php ] -> { Active Session Present? } | +----( No Check
)--------------------+----( Yes Check ) v v [ Render Dark Auth Form ] [ Display Portal Home
Panel ] | | [ Execute Password Validation ] v | [ Query Approved Clinic Nodes ] v | [ Set
Session Variables ] ---> [ Redirect ] v [ Select Target Action & Date ] | v [ Post SQL Booking
Transaction ] | v [ Append Row to History Matrix ] | v [ END ]
```

4.3 Process State Diagram Matrix

```
+------------------------+ Central Admin Clears +-----------------------+ | HOSPITAL NODE STATE
| -----------------------------------> | HOSPITAL NODE STATE | | STATUS: "PENDING" | | STATUS:
"APPROVED" | +------------------------+ +------------------------+ | | System Rejected by Admin
v +------------------------+ | HOSPITAL NODE STATE | | STATUS: "REJECTED" |
+------------------------+ +------------------------+ Hospital Accepts Slot
+------------------------+ | APPOINTMENT LIFE-CYCLE | -----------------------------------> |
APPOINTMENT LIFE-CYCLE | | STATUS: "PENDING" | | STATUS: "APPROVED" | +------------------------+
+------------------------+ | | | Out-of-Capacity Reject Flag | Clinician Posts Result v v
+------------------------+ +------------------------+ | APPOINTMENT LIFE-CYCLE | | APPOINTMENT
LIFE-CYCLE | | STATUS: "REJECTED" | | STATUS: "RESULT FILLED"| +------------------------+
+------------------------+
```

4.4 Relational Database Schematics

The relational layer uses indexed mobile tokens as lookups, with transactional integrity enforced via foreign keys mapped inside the InnoDB storage subsystem.

```
TABLE [ patient ] { id INT PRIMARY KEY AUTO_INCREMENT mobile_number VARCHAR(15) UNIQUE INDEX
full_name VARCHAR(100) password_hash VARCHAR(255) physical_address TEXT location_details
VARCHAR(100) } TABLE [ hospital ] { id INT PRIMARY KEY AUTO_INCREMENT hospital_name VARCHAR(150)
address TEXT location_details VARCHAR(100) functional_state ENUM('PENDING', 'APPROVED',
'REJECTED') } TABLE [ appointment_registry ] { id INT PRIMARY KEY AUTO_INCREMENT patient_id INT
FOREIGN KEY REFERENCES patient(id) hospital_id INT FOREIGN KEY REFERENCES hospital(id)
appointment_type ENUM('TEST', 'VACCINATION') scheduled_date DATE request_status ENUM('PENDING',
'APPROVED', 'REJECTED') execution_result VARCHAR(100) }
```

5.0 SCREEN SHOTS (INTERFACE MOCKUP INTERNALS)

5.1 Patient Interface View Engine // Dark and Twisty Vibe

```
================================================================================== ORS CORE PORTAL // SYSTEM ACCESS: PATIENT DASHBOARD [ Disconnect ] ================================================================================== Identity Descriptor Token: Meredith Grey // System Log Session Verified Active [+] Schedule New Diagnostics Entry Frame ---------------------------------------------------------------------------------- * Select Clinical Network Hub Node: [ Seattle Grace General Hospital (#001) ] * Target Medical Evaluation Track: [ COVID-19 Diagnostic Screening Test ] * Preferred Execution Calendar Date: [ 2026-06-02 ] [ SUBMIT AND AUTHORIZE APPLICATION ] [-] Personal Diagnostic Ledger History & Lab Outcomes ---------------------------------------------------------------------------------- ID Facility Node Service Date Status Outcome ---------------------------------------------------------------------------------- #0042 Seattle Grace General TEST 2026-05-20 APPROVED COVID-19 NEGATIVE #0071 Mercer Island Clinic VACCINE 2026-05-28 PENDING LAB PENDING ==================================================================================
```

5.2 Hospital Workstation Triage System // Bright & Shiny Vibe

```
================================================================================== CLINICAL TRIAGE COMMAND MATRIX // ENVIRONMENT WORKSPACE STATE: STERILE LIGHT ================================================================================== Operational Tracking Node Context Identifier: Seattle Grace General (APPROVED) Active System Sorting Ledger Queue: ---------------------------------------------------------------------------------- Case ID Patient Profile Details Track Date State Actions Map ---------------------------------------------------------------------------------- #AR-302 Alex Karev TEST 2026-05-29 PENDING [Approve] [Reject] Cell Index: 079432109 #AR-114 Cristina Yang VACCINE 2026-05-28 APPROVED [Update Outcome] Cell Index: 071192834 Current: UNRESOLVED ==================================================================================
```

6.0 SOURCE CODE MATRIX (PRODUCTION BACKBONE)

6.1 Secure Relational Setup Blueprint (`schema.sql`)

```sql
-- ========================================================================
-- FILE: schema.sql
-- DEVELOPMENT STRATEGY: Enforces absolute data boundaries inside InnoDB engine.
-- ========================================================================

CREATE DATABASE IF NOT EXISTS ors_covid_db;
USE ors_covid_db;

CREATE TABLE IF NOT EXISTS patient (
    id INT AUTO_INCREMENT PRIMARY KEY,
    mobile_number VARCHAR(15) NOT NULL UNIQUE,
    full_name VARCHAR(100) NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    physical_address TEXT NOT NULL,
    location_details VARCHAR(100) NOT NULL,
    INDEX(mobile_number)
) ENGINE=InnoDB;

CREATE TABLE IF NOT EXISTS hospital (
    id INT AUTO_INCREMENT PRIMARY KEY,
    hospital_name VARCHAR(150) NOT NULL,
    address TEXT NOT NULL,
    location_details VARCHAR(100) NOT NULL,
    functional_status ENUM('PENDING', 'APPROVED', 'REJECTED') DEFAULT 'PENDING'
) ENGINE=InnoDB;

CREATE TABLE IF NOT EXISTS appointment_registry (
    id INT AUTO_INCREMENT PRIMARY KEY,
    patient_id INT NOT NULL,
    hospital_id INT NOT NULL,
    appointment_type ENUM('TEST', 'VACCINATION') NOT NULL,
    scheduled_date DATE NOT NULL,
    request_status ENUM('PENDING', 'APPROVED', 'REJECTED') DEFAULT 'PENDING',
    execution_result VARCHAR(100) DEFAULT 'PENDING LAB UPDATE',
    FOREIGN KEY (patient_id) REFERENCES patient(id) ON DELETE CASCADE,
    FOREIGN KEY (hospital_id) REFERENCES hospital(id) ON DELETE CASCADE
) ENGINE=InnoDB;
```

6.2 Core Pipeline Initialization Link (`db_connect.php`)

```php
<?php
/**
 * FILE: db_connect.php
 * REFACTORING ROADMAP: Implements optimized single-connection parameters.
 * HARDWARE STRATEGY: Tailored to prevent connection spikes on limited physical memory
architectures.
 */

define('DB_SERVER', 'localhost');
define('DB_USERNAME', 'ors_root');
define('DB_PASSWORD', 'SecureDarkPass2026!');
define('DB_NAME', 'ors_covid_db');

// Establish raw socket connection string mapping
$conn = mysqli_connect(DB_SERVER, DB_USERNAME, DB_PASSWORD, DB_NAME);

// Validate active transmission stream before allowing page code blocks to fire
if (!$conn) {
    die("CRITICAL ANOMALY: Database link infrastructure failed: " . mysqli_connect_error());
}

mysqli_set_charset($conn, "utf8");
?>
```

6.3 Administrative Dynamic Data Export Routine (`admin_export_xls.php`)

```php
<?php
/**
 * FILE: admin_export_xls.php
 * STRATEGY: Dynamic streaming layer using low-overhead tab spacing structures (        ).
 * MEMORY PROTECTION: Runs row extractions step-by-step, keeping memory usage clean inside 128 MB
RAM limits.
 */
session_start();
require_once('db_connect.php');

// Security Authorization Filter
if (!isset($_SESSION['admin_authenticated'])) {
    die("SECURITY EXCEPTION: Operation terminated due to missing verification signatures.");
}

$start_date = mysqli_real_escape_string($conn, $_GET['start']);
$end_date   = mysqli_real_escape_string($conn, $_GET['end']);

$query = "SELECT p.full_name, p.mobile_number, a.appointment_type, a.scheduled_date,
a.execution_result
          FROM appointment_registry a
          INNER JOIN patient p ON a.patient_id = p.id
          WHERE a.scheduled_date BETWEEN '$start_date' AND '$end_date'
          ORDER BY a.scheduled_date ASC";

$result = mysqli_query($conn, $query);

// Send direct browser control header declarations
$filename = "ORS_DATA_DUMP_" . $start_date . "_TO_" . $end_date . ".xls";
header("Content-Type: application/vnd.ms-excel");
header("Content-Disposition: attachment; filename="$filename"");
header("Cache-Control: max-age=0");

// Stream column spacing layout lines out to target files
echo "Patient Name   Mobile Index    Evaluation Track        Target Execution Date   Diagnostic
Status
";

// Fetch and clear records sequentially without creating large memory buffers
while ($row = mysqli_fetch_assoc($result)) {
    echo $row['full_name'] . "       ";
    echo $row['mobile_number'] . "       ";
    echo $row['appointment_type'] . "       ";
    echo $row['scheduled_date'] . "       ";
    echo $row['execution_result'] . "
";
}

mysqli_free_result($result);
mysqli_close($conn);
exit;
?>
```

6.4 Perl Background Asset Scoping Daemon (`check_inventory.pl`)

```perl
#!/usr/bin/perl
# ==========================================================================
# FILE: check_inventory.pl
# PURPOSE: Background parsing routine intended for automated system cron configurations.
# ==========================================================================
use strict;
use warnings;
use DBI;

my $dsn = "DBI:mysql:database=ors_covid_db;host=localhost";
my $db_user = "ors_root";
my $db_pass = "SecureDarkPass2026!";

my $dbh = DBI->connect($dsn, $db_user, $db_pass, { RaiseError => 1, AutoCommit => 1 })
    or die "CRITICAL EXCEPTION: " . DBI->errstr;

my $sth = $dbh->prepare("SELECT vaccine_name FROM vaccine_inventory WHERE current_availability =
0");
$sth->execute();

print "[DAEMON ALERT RUN LOG]: COMMENCING LOGISTICAL INVENTORY SWEEP...
";
while (my @row = $sth->fetchrow_array()) {
    print ">> INVENTORY ALERT: Biological Batch Asset Category [" . $row[0] . "] is OUT OF STOCK.
";
}

$sth->finish();
$dbh->disconnect();
exit;
```

7.0 USER GUIDE (OPERATIONAL MANUAL)

The interface layout dynamically alters its structural styling components based on the credential level matching the logged-in user context.

7.1 Patient Workspace Operations

- **Profile Configuration:** Launch path `patient_portal.php`. Use the registration container block on the right to input name details, home location parameters, and your cell index, which establishes your permanent account data identifier.
- **Filing a Triage Application Request:** Once authenticated within the dark interface sanctuary, locate the panel titled **"File Diagnostics Appointment Application"**. Choose an available clinical facility location, pick your care track, and submit. The record will instantly populate inside your secure historical progress table.

7.2 Clinical Workstation Desk Operations

- **Node Activation:** Access tracking module path `hospital_portal.php`. Pick your cleared hospital site from the dropdown and authorize the session block to open your active triage dashboard interface.
- **Queue Triage Processing:** Locate rows marked with standard `PENDING` flags. Click **Approve** to allocate local floor capacity, or **Reject** to dismiss the application. For active entries, apply final outcome metrics via the dropdown controls.

7.3 Administrative Governance Terminal

- **Facility Onboarding:** Access the system layout inside `admin_portal.php` to verify newly registered hospital slots and clear them for public booking operations.
- **Spreadsheet Report Extraction:** Navigate to the Excel extraction control box tool, supply start and end date bounds, and execute. The engine compiles matching entries line-by-line and triggers an immediate local spreadsheet file download.

8.0 DEVELOPER'S GUIDE (STRUCTURAL MODULE MATRICES)

8.1 Module Architecture Logs

The software design eliminates runtime overhead from tracking calculations, using server-side processing to keep memory needs light and preserve performance under the targeted legacy hardware environments.

- **`db_connect.php` (Data Link Module):** Centralized link engine. Relies on procedural MySQLi structures to reduce connection footprints on legacy computers. Avoid object abstractions.

- **`patient_portal.php` (Sanctuary Interface Module):** Locked inside dark charcoal structural parameters (`#090d16`). Uses single-page conditional state checks based on hidden form post markers. Passwords use secure cryptographic hash routing keys.
- **`admin_export_xls.php` (Low-Footprint Streaming Engine):** Emits direct transmission controls (`application/vnd.ms-excel`) and transfers data blocks row-by-row onto system outputs. This prevents heap expansion errors, protecting system stability inside the 128 MB RAM baseline limit.

8.2 Maintenance and Validation Matrix Checklist

Execution Bit	Operational Maintenance Task Verification Criteria	Status Sign-Off Flag
SEC-AUD-01	Run string sanitization structures across all active input vectors to block injection scripts.	AUDIT VERIFIED
MEM-AUD-02	Verify report export modules stream rows sequentially without creating large intermediate server-side arrays.	AUDIT VERIFIED
UI-AUD-03	Audit theme styles to ensure complete distinction between dark patient screens and high-exposure medical workspace layouts.	AUDIT VERIFIED

Chief Systems Administrator
Central Data Infrastructure Team Sign-Off

Lead Security Auditor
Compliance Deployment Matrix Assurance Verified